External Services, Integrations and Monthly Cost

What the CRM needs from outside services, why, what each costs, and what code is still to write | 18 October 2026

AED 554	AED 1,923-6,401	~AED 437	AED 15-40k
Our monthly running cost at launch, incl. 5% VAT (AED 528 + AED 26). Integrations still to build add AED 0	At 1 million users, incl. VAT, with email, files, database and logs growth (section E)	WhatsApp, paid by each company directly to Meta (example: 5,000 leads a month)	One-off security penetration test before go-live

For every outside service this list shows why the CRM needs it, which account is needed and whose it is, which API keys are needed, whether integration code still has to be written, and its monthly cost. Costs are the approved figures from the Technology Plan v1.2 (11 October 2026): AWS UAE region list prices, 1-year commitments for servers and database, 1 USD = 3.6725 AED, launch size of one brokerage with about 50 users and 5,000 new leads a month. Amounts are before VAT unless stated.

Group	Meaning	Items	AED / month
A. Code built: accounts, keys and cloud setup only	Built and tested with local stand-ins. Going live needs the accounts, keys and cloud setup.	20	433
B. Needs integration code	Account and keys are needed, and new code has to be written to connect the service.	16	95
C. Needs a provider choice first	SMS and calls / IVR: a provider must be chosen (recommendation included).	2	0
Total before VAT		38	**528**

Accounts we (the CRM company) must open: AWS, Cloudflare, domain registrar, Stripe, Meta for Developers, Google Cloud, Microsoft Entra (Azure), Firebase, Sentry, GitHub, Better Stack, Zoom Marketplace and Calendly developer apps; later Google Ads API, Microsoft Advertising, LinkedIn / Snapchat / TikTok developer apps, an SMS provider and a telephony provider.
Accounts each customer company brings (its own): Facebook page, WhatsApp Business account and number, Property Finder / Bayut / Dubizzle accounts, Google Ads and Microsoft Ads accounts, Zoom and Calendly plans, and each agent's Google or Microsoft account for calendar and meeting links.

- Real keys and passwords are never put in code; they are stored in AWS SSM Parameter Store.
- Not possible in a web app: **Disable Screenshots**. Browsers do not let a website block screenshots; only a mobile app could.

A. Code built: needs only accounts, keys and cloud setup

#	Service (provider / cloud)	Why we need it	Account needed (whose)	API keys / credentials	Integration code	AED / month
A1	**App servers**: AWS EC2, 2 x t4g.small (UAE region)	The computers that run the CRM. Two of them in two data centres, so one can fail without users noticing. Data stays in the UAE.	Our company AWS account	Server access roles (IAM)	**Built** (app). **Needed**: deployment scripts (Terraform).	**73**
A2	**Database**: AWS RDS PostgreSQL, Multi-AZ	Keeps every lead, call, note, project, invoice and attendance record. A live standby copy takes over automatically if the main one fails.	Our AWS account	Database passwords (kept in AWS SSM)	**Built**	**272**
A3	**Database storage and backups**: AWS	Disk space plus daily backups kept 14 days, so any moment in that window can be restored.	Our AWS account	-	**Built**	**18**
A4	**Internet addresses**: AWS public IPv4 (2)	Lets the servers reach WhatsApp, Meta, Google and the property portals.	Our AWS account	-	**Setup** with deployment	**26**
A5	**File storage**: AWS S3	Holds brochures, floor plans, images, invoice PDFs, attendance photos and Excel exports.	Our AWS account	Bucket and access role	**Built**	**7**
A6	**Monitoring**: AWS CloudWatch	Watches server health and raises an alarm if something looks wrong.	Our AWS account	-	**Setup**: alarms are set in the deployment scripts	**11**
A7	**Test environment**: AWS EC2 Spot, office hours	A separate copy for testing changes before staff see them (required by the brief).	Our AWS account	-	**Setup** with deployment	**11**
A8	**Email**: Amazon SES (+ SNS)	Sends invites, reminders, meeting confirmations and reports. Receives portal lead emails (Property Finder, Bayut, Dubizzle).	Our AWS account; our sending domain verified	Domain DNS records; leave SES sandbox	**Built**	**11**
A9	**Login**: Amazon Cognito	Secure sign-in for the CRM and our owner console; invites new company Admins. Free up to 10,000 active users a month.	Our AWS account	User pool IDs, app client IDs	**Built** (2FA: see B12)	0
A10	**Secrets store**: AWS SSM Parameter Store	Keeps passwords and integration keys out of the code and out of the database.	Our AWS account	-	**Built**	0
A11	**Security and speed**: Cloudflare Free	Stands in front of the CRM: blocks attacks, SSL padlock, DNS, fast images; Turnstile bot check on public sign-up.	Our Cloudflare account	Turnstile secret key	**Built**	0
A12	**Domain name** (.com or .ae)	The web address of the CRM.	Our registrar account	-	-	**4**
A13	**Card payments**: Stripe	Takes subscription payments by card (14-day trial, plans, invoices). Card details go only to Stripe.	Our Stripe UAE account (trade licence)	Secret, publishable and webhook keys	**Built**	0 *
A14	**Facebook / Instagram lead ads**: Meta for Developers app	Ad leads arrive automatically; the Meta check finds and recovers missing leads.	Our Meta app (business verification + App Review); each company connects its page	App ID, app secret, verify token	**Built**	0

#	Service (provider / cloud)	Why we need it	Account needed (whose)	API keys / credentials	Integration code	AED / month
A15	**WhatsApp**: WhatsApp Business Platform (Meta Cloud API)	WhatsApp chats on the lead timeline, templates and campaigns, straight with Meta (no third-party inbox).	Each company's own WhatsApp Business account and number	Phone number ID and token, entered by the company	**Built**	0 **
A16	**Property Finder, Bayut, Dubizzle**	Portal leads arrive in the CRM without retyping.	Each company's own portal account	Webhook secret or CRM lead email address	**Built**	0
A17	**General webhook** (built into the CRM)	One secure link any system can send leads to: website, landing pages, Google Sheets.	None	Created in the CRM	**Built**	0
A18	**PDFs and Excel**: Gotenberg, ExcelJS (open source)	Invoice and purchase order PDFs and Excel exports, made on our own servers.	None	-	**Built**	0
A19	**Code storage and delivery**: GitHub Free + Actions	Keeps the source code (owned by the company) and releases tested updates automatically.	Company GitHub account	Deployment access to AWS	**Built** (code). **Setup**: release pipeline.	0
A20	**Uptime alerts**: Better Stack Free	Checks every few minutes that the CRM is reachable and alerts the team if not.	Our Better Stack account	-	No code needed	0

* Stripe has no monthly fee; it takes about 2.9% + AED 1 per card payment from revenue. ** Each company pays Meta for its own WhatsApp messages; replies within 24 hours of a customer message are free.

B. Needs integration code

#	Service (provider / cloud)	Why we need it	Account needed (whose)	API keys / credentials	Integration code	AED / month
B1	**Error tracking**: Sentry Team	Tells developers instantly when something breaks, with details to fix it fast.	Our Sentry account	Project key (DSN)	**Small change**: connect Sentry to the API and web app.	**95**
B2	**Google Calendar + Google Meet** (Google Workspace)	Meet links created automatically for online meetings (replaces Zoom) and two-way calendar sync for agents.	Our Google Cloud project (verified consent screen); agents use their own accounts	OAuth client ID and secret	**Needed**: "connect Google account" sign-in. Sync and Meet-link code exist.	0
B3	**Outlook Calendar + Teams** (Microsoft 365)	Same as B2 for agents on Microsoft: Teams links and calendar sync.	Our Microsoft Entra (Azure) app; agents use their own accounts	Client ID and secret	**Needed**: "connect Microsoft account" sign-in. Sync and Teams-link code exist.	0
B4	**Gmail** (Gmail API)	Integration card: send and receive email through a company Gmail account.	Our Google Cloud project; Google security review for Gmail access	OAuth client ID and secret	**Needed**: full connector.	0 ***
B5	**Google Ads Lead Form**	Leads from Google Ads lead form extensions arrive automatically.	Each company's Google Ads account	Webhook key set in Google Ads	**Needed**: connector.	0
B6	**Google Campaign** (Google Ads API)	Ad spend and results imported daily for cost-per-lead reports (today spend is typed in).	Google Ads API developer token (approved by Google); company connects its Ads account	Developer token, OAuth client	**Needed**: connector and daily import.	0
B7	**Microsoft Advertising** (API)	Leads and ad spend from Microsoft Ads.	Microsoft Advertising developer access; company connects its account	Developer token, Entra app	**Needed**: connector.	0
B8	**LinkedIn, Snapchat, TikTok** lead ads (APIs)	Phase 2 lead sources in the SRS: ad leads arrive with campaign and form details.	Developer app with each platform, approved for lead access (LinkedIn approval takes weeks)	App ID and secret each	**Needed**: one connector per platform.	0
B9	**Skyloov**	Property portal leads (integration card).	API or lead-feed access from Skyloov	From Skyloov	**Needed**: connector (or Skyloov posts to the general webhook).	To confirm
B10	**Phone push notifications**: Firebase Cloud Messaging	Pushes new-lead alerts and reminders to agents' phones. Today alerts show inside the CRM only.	Our Firebase project	Service account key	**Needed**: device registration and sending.	0
B11	**Maps**: Google Maps Platform (free allowance)	Shows project and unit locations on a map.	Our Google Cloud project	Maps API key	**Needed**: map view on projects and units.	0
B12	**Two-factor sign-in (2FA)**: Amazon Cognito	Extra login protection with a code from an authenticator app ("Require 2FA" switch).	Our AWS account (A9)	Cognito MFA settings	**Small change**: apply the saved switch to sign-in.	0
B13	**SMS** (provider not chosen)	SMS templates and SMS sending.	Account with chosen provider; UAE sender ID registration	Provider API key	**Needed**: SMS channel.	To be quoted

#	Service (provider / cloud)	Why we need it	Account needed (whose)	API keys / credentials	Integration code	AED / month
B14	**Calls and IVR** (provider not chosen)	IVR (number, greeting, menu, routing), call notifications and call detection.	Account with chosen provider; UAE phone number	Provider API key	**Needed**: IVR and call integration.	To be quoted
B15	**Zoom** (kept: stakeholder decision 18 Oct 2026)	Zoom meeting links created automatically from the agent's connected Zoom account (today agents paste a link).	Our Zoom Marketplace app; each brokerage's own paid Zoom plan	Client ID and secret	**Needed**: "connect Zoom account" and meeting creation.	0 ****
B16	**Calendly** (kept: stakeholder decision 18 Oct 2026)	Agents send their Calendly booking link; bookings and changes in Calendly update the appointment on the lead.	Our Calendly app; each brokerage's own paid Calendly plan	Client ID and secret, webhook key	**Needed**: "connect Calendly" and booking sync.	0 ****

*** The Gmail API itself is free; Google's security review for Gmail access may have a fee, to be confirmed when we apply.

C. Needs a provider choice first

#	Service (provider / cloud)	Why we need it	Account needed (whose)	API keys / credentials	Integration code	AED / month
C1	**SMS provider** (B13)	Choose before building. UAE rules: the sender name must be registered with TDRA through e& or du.	Recommended (low cost): a UAE aggregator connected directly to e& and du, e.g. Unifonic or D7 Networks; get 2-3 quotes	Provider API key	About AED 0.06-0.10 per SMS (vendor quotes) vs about AED 0.43 with Twilio	Per SMS
C2	**Calls and IVR provider** (B14)	Choose before building. UAE rules: business calling must use a TDRA-licensed network.	Recommended: each brokerage's own e& or du business cloud phone line (the licensed route); confirm it offers an API	Provider API key	Quote from e& / du; paid by each brokerage	To be quoted

D. Grand total: monthly cost to run the entire CRM (launch)

Service	Ref	AED / month
App servers (AWS EC2)	A1	73
Database (AWS RDS PostgreSQL, Multi-AZ)	A2	272
Database storage and backups	A3	18
Internet addresses (AWS IPv4)	A4	26
File storage (AWS S3)	A5	7
Monitoring (AWS CloudWatch)	A6	11
Test environment (AWS EC2 Spot)	A7	11
Email (Amazon SES)	A8	11
Error tracking (Sentry Team)	B1	95
Domain name	A12	4
All other services in this list (Cognito, SSM, Cloudflare, Stripe (monthly), Meta, portals, webhook, Gotenberg, GitHub, Better Stack, Google / Microsoft calendars and Meet / Teams, Gmail API, Google Ads, Microsoft Ads, LinkedIn / Snapchat / TikTok, Firebase, Google Maps (free allowance), 2FA)	A, B	0
Total before VAT		**528**
UAE VAT 5%		26
GRAND TOTAL per month (our cost to run the CRM)		**AED 554**

Not in our monthly cost	Who pays	Amount
WhatsApp messages (example: welcome message to 5,000 new leads plus follow-ups)	Each company, directly to Meta	about AED 437 / month incl. VAT
Card payment fees (Stripe UAE)	Taken from our subscription revenue	about 2.9% + AED 1 per payment
SMS (B13, C1)	To decide with the provider choice	about AED 0.06-0.10 per SMS
Calls / IVR (B14, C2)	Each brokerage, on its own line	To be quoted
Skyloov access (B9)	To confirm with Skyloov	To confirm
Zoom and Calendly (B15, B16)	Each brokerage, on its own plans	about AED 52-62 / user, AED 37-44 / seat
Advertising spend on Meta, Google, Microsoft, LinkedIn, Snapchat, TikTok	Each company	Their own budget

One-off costs (not monthly)

- Security penetration test before go-live: about AED 15,000-40,000, usually once a year.
- UAE SMS sender ID registration, only if SMS is added.
- Native iPhone / Android apps, only if needed: Apple AED 364 a year, Google AED 92 once.

Notes

- Servers and database are on 1-year commitments; paying month by month would add about AED 150 a month.
- At about 500 users (10 brokerages) the same design costs about AED 1,508 a month incl. VAT (Technology Plan v1.2, section 6).
- AWS amounts are estimates from public UAE list prices; confirm in the AWS Pricing Calculator before signing.

E. One million users: services that grow with use

Normal use = Technology Plan v1.2 design (phone push first, email only when needed). **Heavy use** = about 4 emails a month to every user and much more stored data. Volumes are estimates to adjust once real usage is known.

Service	Price	Normal	AED	Heavy	AED
Emails sent (Amazon SES)	USD 0.10 / 1,000	200,000	73	4 million	1,469
Email attachments	USD 0.12 / GB	4 GB	2	80 GB	35
Portal lead emails received (SES)	USD 0.10 / 1,000	100,000	37	200,000	73
Property photos and brochures (Cloudflare R2, free to view)	USD 0.015 / GB-month	2 TB	110	10 TB	551
Photo requests not served from cache (R2)	USD 0.36 / M reads, 4.50 / M writes	within free tier	0	50 M reads	69
Private files: invoices, attendance photos, attachments, exports (AWS S3 UAE)	about USD 0.025 / GB-month *	1 TB	92	5 TB	459
Private file downloads (AWS)	about USD 0.12 / GB *	100 GB	44	1 TB	441
Database disk growth: leads, timelines, audit log, history (x2 for standby copy)	about USD 0.097 / GB-month *	500 GB	355	2 TB	1,422
Server logs (AWS CloudWatch)	USD 0.50 / GB, first 5 GB free	50 GB	83	300 GB	542
Phone push notifications (Firebase)	Free	unlimited	0	unlimited	0
Usage costs per month			**796**		**5,061**

Grand total at 1 million users	Normal	Heavy
Servers, own database with standby, self-hosted login, free maps, backups (Technology Plan v1.2, section 14)	1,035	1,035
Usage costs (table above)	796	5,061
UAE VAT 5%	92	305
GRAND TOTAL per month	**AED 1,923**	**AED 6,401**

Costs avoided by design (not in the total)

Expensive option	Cost at 1 million users	What we use instead
Amazon Cognito login above 10,000 active users	about AED 54,000 / month (Technology Plan v1.2)	Self-hosted login (Keycloak): AED 0
Google Maps (USD 7 per 1,000 map loads after 10,000 free)	about AED 7,455 / month at 300,000 loads	OpenStreetMap maps: AED 0
SMS for alerts	about AED 600-1,000 per 10,000 SMS with a local provider (AED 4,319 with Twilio)	Free phone push, then WhatsApp or email. SMS only if the stakeholder chooses it.
WhatsApp to every user	about AED 55,000 per message to 1 million people	Paid by each company directly to Meta, never by us

- Cost per user stays tiny: about AED 0.002 (normal) to 0.006 (heavy) per user a month.
- Biggest swings: email (each extra 1 million emails adds about AED 367) and stored data. Building phone push (B10) and archiving old data keep both down.
- * AWS UAE prices for S3 and database disk are estimates; confirm in the AWS Pricing Calculator. Other prices: official SES, R2, CloudWatch, Google Maps and Twilio pricing pages, October 2026.